

Homework — 2.1.1 Thinking Abstractly — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. (a) [3] Completed table:

Detail of the city	Keep or Discard?
Locations of junctions and delivery addresses	Keep (they become the nodes of the graph)
Typical travel time along each road	Keep (the edge weights — needed to find the <i>quickest</i> route)
One-way restrictions on roads	Keep (they determine which routes are actually possible)
Street names	Discard (no effect on how long a road takes to travel)
Colours of the buildings	Discard (irrelevant to routing)
Texture of each road's surface	Discard (its effect is already captured in the travel time)

Mark: 2 correct rows per mark (6 rows → 3 marks). The bracketed reasons are not required for the marks.

(b) [1] Any one: the model is **simpler**, so it is faster to build / needs less storage and processing / is easier to understand, test and maintain — while still solving *this* problem.

2. [3] Completed paragraph, in order of the blanks:

1. **unnecessary**
2. **purpose**
3. **higher**
4. **controls**
5. **danger**
6. **decomposition**

Unused distractors: lower, abstraction, hardware.

Mark: 2 correct blanks per mark (6 blanks → 3 marks).

3. (a) [1] The skill is **decomposition** (thinking procedurally) — breaking a problem down into smaller sub-problems that can be solved separately. (*Abstraction, by contrast, removes/hides unnecessary detail — a common trap from the spec.*)

(b) [1] Any one genuine abstraction, e.g. representing the board as a **2D array** of values rather than modelling physical pieces; hiding how a move is validated behind a simple "is this move legal?" interface; ignoring irrelevant details such as the material or look of the pieces when checking rules.

4. (a) [2] 1 mark each, any two, e.g.: a "crash" has **no real consequences/danger**; motion and G-forces are absent or only approximated; the world outside the cockpit is **computer-generated graphics**, not real

terrain/weather; weather and equipment failures are generated on demand by software; no real fuel is used / no wear on a real aircraft.

(b) [1] The model **keeps the details that matter** for training (the controls, instrument readings and how the aircraft responds) while removing irrelevant or dangerous ones — the abstraction only needs to match reality for its **purpose**, so pilots practise the same decisions and skills safely and cheaply.

5. [2] 1 mark for a sensible *keep*, e.g. the visitor's place/position in the queue, arrival/join time, party size, ride eligibility (height). 1 mark for a discarded detail **with justification**, e.g. the visitor's favourite colour / what they had for lunch — it plays no part in queue position or wait time, so it is irrelevant to this model. Must apply to *this* scenario.

Part B (6 marks)

6. [2] 1 mark: the PC **holds the address of the next instruction** to be fetched. 1 mark: this address is copied to the MAR at the start of fetch, and the PC is then **incremented** (or updated by a jump) so the cycle moves through the program in order.

7. [2] 1 mark + 1 mark, any one developed advantage: SSDs have **no moving parts**, so they are faster (lower access/seek time) **[1]**; also more durable/shock-resistant, silent and lower power **[1]**. (*Accept any one valid advantage stated clearly for both marks.*)

8. [2] 1 mark: pipelining is fetching/decoding/executing **overlapped** — while one instruction is executed, the next is decoded and the one after is fetched. 1 mark: benefit — more instructions are completed per unit time / better use of the CPU / higher throughput.

Total: 14 + 6 = 20 marks